Research Proposal: Assignment Brief (Individual Product 1)

Foundations of Clinical Research · Level 1: From Clinical Question to Publication

Organisation: Neudata (*#ClearDataClearImpact*)

Assessment type: Individual assignment. Each part is modelled live in Sessions 1 to 7 and drafted by you as homework. Submitted in Session 8.

Who: every participant writes their own proposal, on a question from their own department. There are no groups.

Output: one written research proposal with a publication plan (about 6 to 8 pages) + 6 slides, submitted by everyone. About 5 volunteers also present their proposal in Session 8 (6 minutes).

The other individual product: your draft manuscript from the course dataset (see `Manuscript_Brief.pdf`)

1. Purpose

Most clinical research starts as a question asked at the bedside. This assignment turns one such question from **your own department** into a realistic, ethical and feasible research proposal: the kind you could take to your research office, an ethics committee or a small-grant panel.

You will not collect data or analyse anything for the proposal. The goal of this product is to **plan a study well** (a clear question, the right design, good data, honest attention to bias and ethics, a realistic timeline) and to **plan from the start how it will be published**: target journal, reporting guideline, authors and a draft abstract. In each session the instructor shows one part of the proposal live, on the running example. You then write the same part for your own question as homework. By Session 7 you have a complete proposal and publication plan, and in Session 8 you submit it.

This is the first of the course's **two individual products**. The second, a draft manuscript written from the results of the simulated course dataset FLUID-ICU, is built in Sessions 5 to 8 and described in `Manuscript_Brief.pdf`. The two products are deliberately different. The proposal is about **your own question** (planning a study that has not been done), and the manuscript is about **data we give you** (reporting a study that has been done).

Throughout the course the instructors use one running example to show what each part looks like: *In adults with septic shock in the ICU, does a restrictive fluid strategy, compared with usual care, reduce 28-day mortality?* **You choose your own question.** It may be from any area of your practice (for example pressure injuries, hand hygiene, delirium, medication errors, ventilator-associated pneumonia, door-to-needle time). Please do not reuse the running example.

A good topic is: a problem you see often, that matters to patients, and that your department could realistically study in 12 to 24 months with its own patients and records.

2. What to produce, Part by Part

Use the Word template `Course/Templates/Research_Proposal_Template.docx` (in the course folder). Its headings match the four Parts below. The template is about 6 to 8 pages when filled. Write in short, plain sentences. Tables and bullet points are welcome. Delete the grey guidance text before submitting.

Part 1: Sessions 1 and 2 (Ask; Find and read)

Section	What we expect
Working title	Built from your PICO: names the population, intervention/exposure, outcome and (ideally) the design, e.g. "Restrictive fluids and 28-day mortality in septic shock: a randomised trial". You will refine it in Part 4.
Problem statement & rationale	What is the problem, how big is it, why does it matter here, what is not yet known (half a page).
PICO	Population, Intervention/exposure, Comparison, Outcome (and Time), one line each.
Research question	One sentence built from the PICO.
FINER check	One line each for Feasible, Interesting, Novel, Ethical and Relevant, plus what you changed because of it.
Aim, objectives, hypothesis	One broad aim; 2 or 3 specific, measurable objectives; one testable hypothesis (if the study compares groups of patients).
5 key references	From your PubMed search after Session 2 (literature review worksheet / literature matrix), stored in Zotero or another reference manager; Vancouver style; at least 3 from the last 10 years.

Part 2: Sessions 3 and 4 (Design; Bias and effect)

Section	What we expect
Study design & justification	Name the design and say why it fits your question (and why a stronger design is or is not possible).
Setting	Type of unit/hospital, period of the study (no real hospital name needed).
Population	Inclusion and exclusion criteria, written so two people would select the same patients.
Exposure/intervention & comparator	What exactly is given or measured, and what it is compared with.
Outcomes	One primary outcome (with how and when it is measured) + 1 to 3 secondary outcomes.
Main biases & how they are reduced	At least 2 specific threats (selection, information, confounding) and the step that reduces each (randomisation, blinding, standard definitions, matching, adjustment).

Part 3: Sessions 5 and 6 (Data; Analyse)

Section	What we expect
Variables table	Outcome, exposure and 3 to 6 confounders/other variables, each with its type (binary, categorical, ordinal, continuous) and unit.
Mini-CRF & data dictionary	A data collection form of 6 to 10 items, and matching data-dictionary rows (name, definition, type, unit/codes, allowed range).
Data management & quality plan	How data will be captured (paper CRF or electronic data capture, e.g. REDCap or KoBoToolbox), coded, checked (range and consistency checks, double entry or spot checks), stored securely with backups, and who can change them (audit trail).
Sample size statement	One or two sentences: expected outcome rates (from the literature), the difference you want to detect, power, significance level and the resulting number. For a descriptive study, give the precision you want instead. Show the tool you used.
Analysis plan in plain language	How each variable will be summarised (mean/SD, median/IQR, n/%), which test or model for the primary comparison, which confounders you will adjust for, and how you will avoid multiple-comparison problems (one pre-specified primary outcome).
Missing-data plan	How you will prevent missing data, how you will report it (how much, for which variables), and what you will do about it (e.g. complete-case analysis, plus multiple imputation as a sensitivity analysis if a statistician is available).

Part 4: Session 7 (Protect and write)

Section	What we expect
Ethics & consent	Is it research, audit or QI? Ethics committee approval route, who consents and how (including patients who lack capacity, if relevant), risks and benefits.
Registration	Where and when the study will be registered (e.g. ClinicalTrials.gov or a WHO ICTRP primary registry for trials, before the first participant; PROSPERO for systematic reviews), or why registration is not required.
Confidentiality	De-identification (study codes), who holds the key, where data are stored, who can access them, how long they are kept.
Feasibility	Timeline (simple Gantt chart), budget headings (no figures needed), people and resources, expected recruitment per month and how you know it is achievable.
Publication plan: target journal & reporting guideline	One target journal (checked with Think. Check. Submit.; scope, open-access fee) and a back-up journal; the reporting guideline for your design (CONSORT, STROBE, STARD, PRISMA, SQUIRE), found via EQUATOR.
Publication plan: authorship & CRediT roles	Who would be an author if the study went ahead in your department (ICMJE criteria), in what order, and each person's CRediT role (e.g. conceptualisation, methodology, data curation, formal analysis, writing of the original draft); who would be acknowledged instead. Roles or job titles are enough; you do not need real names.
Publication plan: working title	The final working title (from Part 1, refined).
Publication plan: structured-abstract skeleton	About 200 words: Background (1 or 2 sentences), Methods (design, setting, participants, primary outcome, analysis), *expected* Results (what you will report, with placeholders, e.g. "28-day mortality was X% vs Y% (RR ..., 95% CI ...)", with no invented numbers), Conclusion (what each possible result would mean).
Dissemination	How results will be fed back to the department (meeting, poster, conference abstract) and to patients if relevant.

Length: about 6 to 8 pages, including the mini-CRF, Gantt chart and abstract skeleton (appendices are fine). Short and clear beats long.

Which session builds what (proposal-builder segments): in each segment the instructor models the part live on the running example and answers questions in the chat. You then draft the same part for your own question at home, before the next session.

Session	Proposal builder	Part
S1	Topic, PICO, research question, FINER check, aim and objectives	1
S2	Search string, 5 references in the literature matrix, problem statement, working title	1
S3	Design and justification, setting, population (inclusion/exclusion)	2
S4	Exposure and comparator, outcomes, main biases and mitigation	2
S5	Variables table, mini-CRF, data-dictionary rows, data-management plan	3
S6	Sample size, analysis plan, missing-data plan	3
S7	Ethics and consent, registration, confidentiality, feasibility, publication plan	4
S8	Submission; about 5 volunteer presentations (no new writing in class)	None

Getting help: ask in the chat during the proposal-builder segment, or email the instructor with a short question between sessions. Instructors may show (with your permission) a good anonymised example from a participant's homework at the start of the next session.

3. Deadlines

All sessions are online, Saturdays 19:30 to 22:30 (Vietnam time).

When	What is due
Before Session 2 (Week 2)	Topic chosen; PICO, research question, FINER check, aim and objectives drafted.
Before Session 3 (Week 3)	Part 1 complete: problem statement, working title and 5 key references.
Before Session 5 (Week 5)	Part 2 complete.
Before Session 7 (Week 7)	Part 3 complete. Start the 6 slides (`Presentation_Guide.pdf`) and think about the target journal and authors.
Before Session 8 (Week 8)	Part 4 complete (ethics, registration, feasibility, publication plan); 6 slides finished. If you would like to present in Session 8, tell the course lead by the Wednesday before Session 8 (about 5 places, first come, first served; the instructor may invite people if there are fewer volunteers).
Session 8 (Week 8, Saturday)	**Everyone submits** the written proposal and the 6 slides (send them before 19:30). About 5 volunteers **present** (6 min + 2 min feedback).
2 weeks after Session 8	Optional revised proposal (Parts 1 to 4, revised after feedback) for a second round of written comments. This is the same deadline as the manuscript response letter.

4. Submission

- **Format:** one Word file per participant, using the template, named `Surname_Proposal.docx` (e.g. `Nguyen_Proposal.docx`), plus your slides as `Surname_slides.pptx` or PDF.

- **Where:** by email to the course lead, or to the shared online course folder announced in Session 1.
- **Include on the cover page:** your name and profession, your department (type of unit is enough) and the working title. (Authorship and CRediT roles for the future study go in the publication plan, Part 4.)
- **Use of AI tools:** you may use AI tools to improve language or to brainstorm, but say so in a short statement at the end ("AI use: ..."), check every fact and reference yourself, and never let AI invent references. You are responsible for every word (Session 7, segment 7.2).
- **Confidentiality:** do not include any real patient data, names or record numbers. Numbers you quote (e.g. "about 40 admissions with septic shock per year") should be approximate unit figures, not extracted records.
- **Originality:** write in your own words and cite every source. Copying text from papers or the internet without quotation marks and a citation is plagiarism. You may discuss your idea with colleagues, but the written proposal must be your own work.

5. Feedback

- **Written proposal (everyone):** marked with the rubric in `Proposal_Rubric.pdf` and returned with written comments within about 3 weeks of submission.
- **Presentation (volunteers only):** 2 minutes of live feedback from the instructor and the class (in the chat), plus a completed feedback form. Presenters are also scored on the presentation (see `Proposal_Rubric.pdf`). Not presenting does not lower your score: non-presenters are scored on the written proposal only, scaled to 100.
- **The marks are formative.** They are there to help you improve the proposal, not to pass or fail anyone.

After the course: strong proposals can be developed into a full protocol and statistical analysis plan (templates in `Course/Templates/`), registered and taken to your research or ethics office. When the data are collected, the manuscript skills from Sessions 7 and 8 apply directly. This course is **Level 1**. **Level 2** (*Doing the analysis*, `Course_Levels/level2_hands_on.md`) teaches you to run the analysis yourself in R, hands-on, and **Level 3** (`Course_Levels/level3_advanced.md`) offers advanced methods for complex data as workshops or consultancy.